

```
!pip install kagglehub
```

```
Requirement already satisfied: kagglehub in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: kagglesdk<1.0,>=0.1.14 in /usr/local/lib/pytho
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/pyt
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.1
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.1
```

```
import kagglehub
```

```
# Download latest version
path = kagglehub.dataset_download(
    "bhanupratapbiswas/customer-behavior-analysis"
)

print("Path to dataset files:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/bhanupratapbiswas/customer-beh
100%|██████████| 9.94M/9.94M [00:00<00:00, 44.9MB/s]Extracting files...
```

```
Path to dataset files: /root/.cache/kagglehub/datasets/bhanupratapbiswas/cust
```

```
import os
```

```
os.listdir(path)
```

```
['ecommerce_customer_data_custom_ratios.csv',
 'ecommerce_customer_data_large.csv']
```

```
import pandas as pd
```

```
df = pd.read_csv(path + "/ecommerce_customer_data_large.csv")
```

```
df.head()
```



	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	
		2023-01-						

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          250000 non-null int64
1   Purchase Date                        250000 non-null object
2   Product Category                    250000 non-null object
3   Product Price                      250000 non-null int64
4   Quantity                          250000 non-null int64
5   Total Purchase Amount              250000 non-null int64
6   Payment Method                     250000 non-null object
7   Customer Age                       250000 non-null int64
8   Returns                           202618 non-null float64
9   Customer Name                      250000 non-null object
10  Age                               250000 non-null int64
11  Gender                            250000 non-null object
12  Churn                             250000 non-null int64
dtypes: float64(1), int64(7), object(5)
memory usage: 24.8+ MB
```

df.isnull().sum()

	0
Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	47382
Customer Name	0
Age	0
Gender	0
Churn	0

dtype: int64

```
df.dropna(inplace=True)

df.drop_duplicates(inplace=True)

print("Cleaning completed")
```

Cleaning completed

```
df.describe()
```

	Customer ID	Product Price	Quantity	Total Purchase Amount	Customer Age
count	202618.000000	202618.000000	202618.000000	202618.000000	202618.000000
mean	25020.140234	254.899066	3.008420	2725.817662	43.81792
std	14412.388674	141.720425	1.414339	1442.268491	15.35606
min	1.000000	10.000000	1.000000	100.000000	18.00000
25%	12599.250000	133.000000	2.000000	1478.000000	30.00000
50%	25018.000000	255.000000	3.000000	2727.000000	44.00000
75%	37444.000000	377.000000	4.000000	3975.000000	57.00000

```
print("MEAN")
print(df.mean(numeric_only=True))

print("\nMEDIAN")
print(df.median(numeric_only=True))

print("\nMODE")
print(df.mode().iloc[0])
```

```
MEAN
Customer ID          25020.140234
Product Price        254.899066
Quantity             3.008420
Total Purchase Amount 2725.817662
Customer Age         43.817923
Returns              0.500824
Age                  43.817923
Churn                0.201088
dtype: float64
```

```
MEDIAN
Customer ID          25018.0
Product Price        255.0
Quantity             3.0
Total Purchase Amount 2727.0
Customer Age         44.0
Returns              1.0
Age                  44.0
Churn                0.0
dtype: float64
```

```
MODE
Customer ID          13805.0
Purchase Date        2020-01-06 22:26:24
Product Category      Clothing
Product Price         290.0
Quantity              4.0
Total Purchase Amount 2174.0
Payment Method        PayPal
Customer Age          47.0
Returns              1.0
Customer Name         Michael Smith
Age                   47.0
Gender                Male
Churn                 0.0
Name: 0, dtype: object
```

```
df.columns
```

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer
Age',
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
```

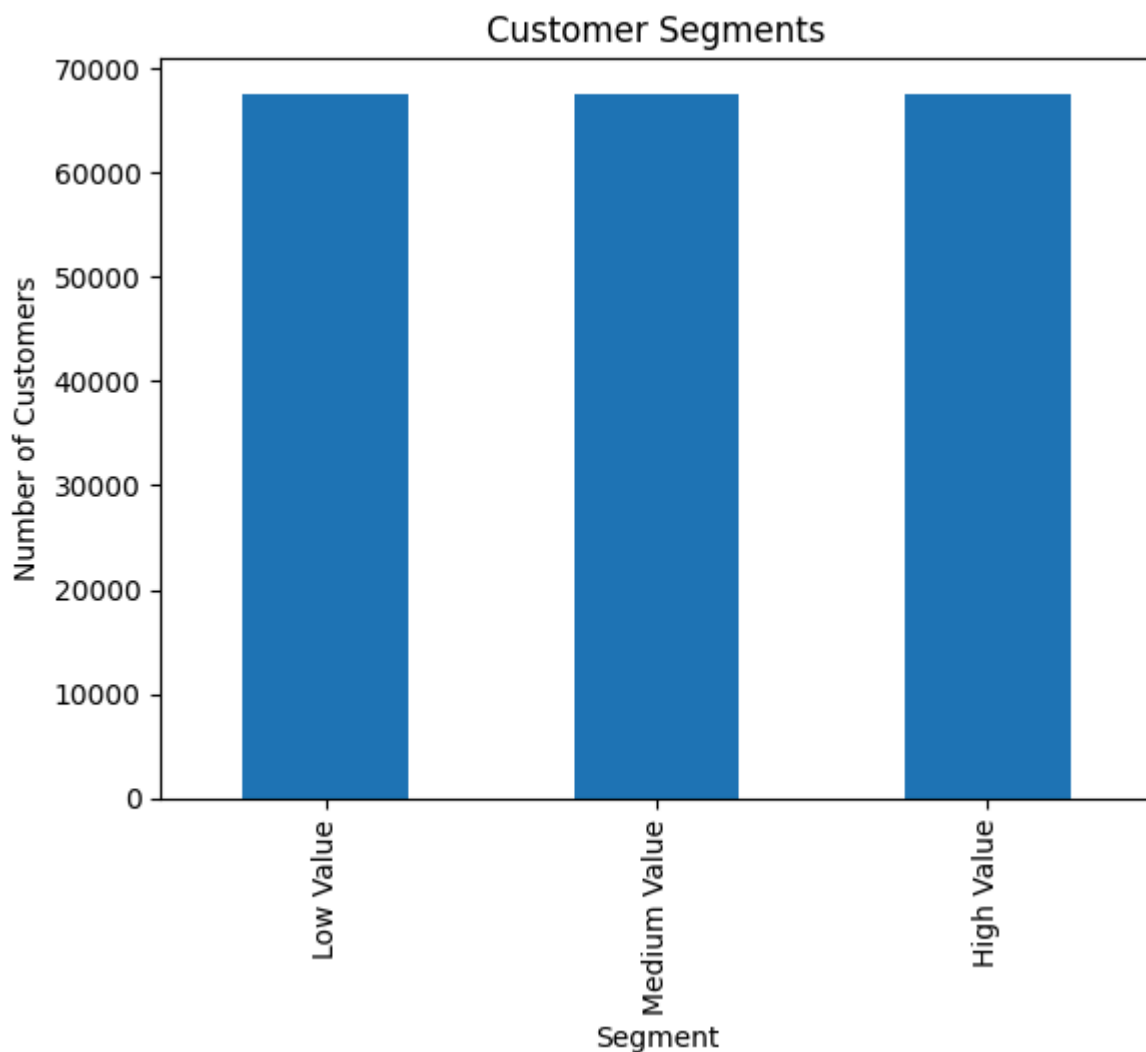
```
df['Price_Per_Item'] = (  
    df['Total Purchase Amount'] / df['Quantity']  
)  
  
df[['Total Purchase Amount', 'Quantity', 'Price_Per_Item']].head()
```

	Total Purchase Amount	Quantity	Price_Per_Item
0	2427	1	2427.000000
1	2448	3	816.000000
2	2345	1	2345.000000
3	937	3	312.333333
4	2598	4	649.500000

```
df['Customer_Segment'] = pd.qcut(  
    df['Total Purchase Amount'],  
    q=3,  
    labels=['Low Value', 'Medium Value', 'High Value']  
)  
  
df[['Total Purchase Amount', 'Customer_Segment']].head()
```

	Total Purchase Amount	Customer_Segment
0	2427	Medium Value
1	2448	Medium Value
2	2345	Medium Value
3	937	Low Value
4	2598	Medium Value

```
import matplotlib.pyplot as plt  
  
df['Customer_Segment'].value_counts().plot(kind='bar')  
  
plt.title("Customer Segments")  
plt.xlabel("Segment")  
plt.ylabel("Number of Customers")  
  
plt.show()
```



```
import pandas as pd
import matplotlib.pyplot as plt
import kagglehub

# Download dataset again
path = kagglehub.dataset_download(
    "bhanupratapbiswas/customer-behavior-analysis"
)

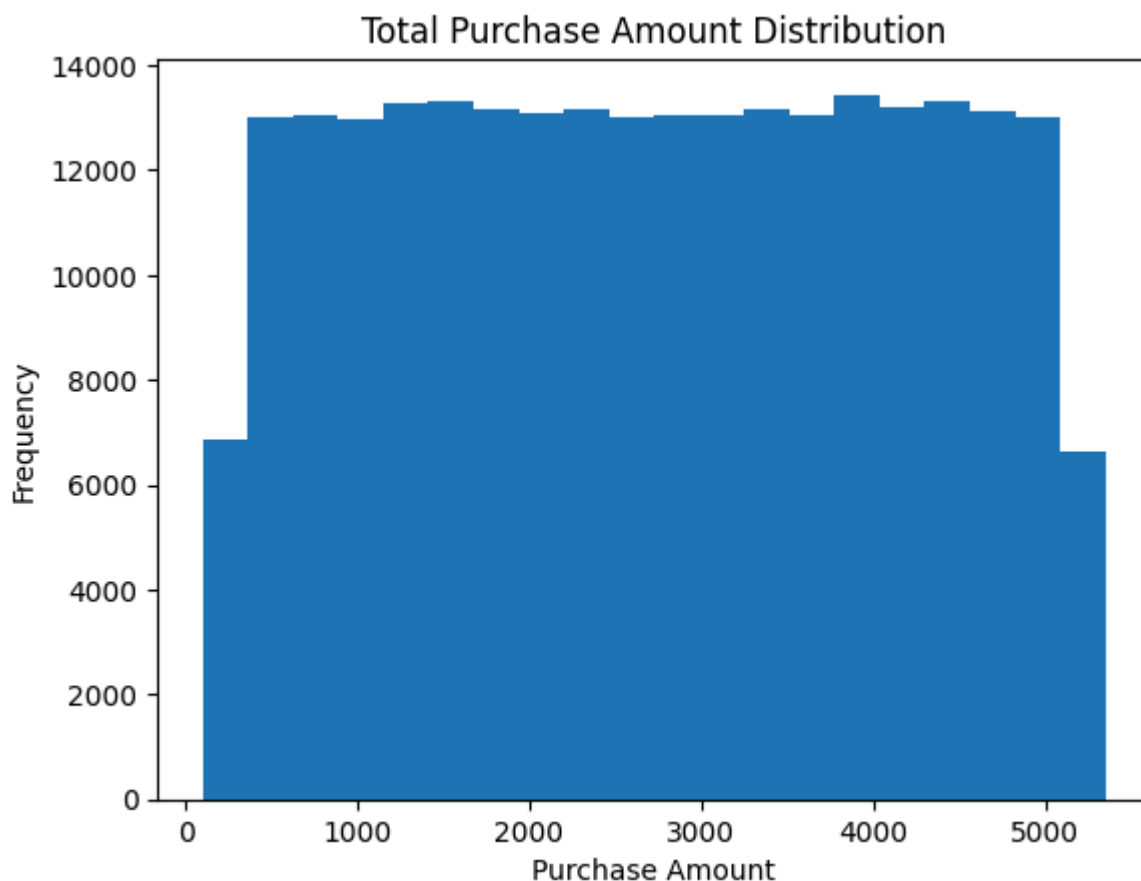
# Load dataset
df = pd.read_csv(path + "/ecommerce_customer_data_large.csv")

# Histogram
plt.hist(df['Total Purchase Amount'], bins=20)

plt.title("Total Purchase Amount Distribution")
plt.xlabel("Purchase Amount")
plt.ylabel("Frequency")

plt.show()
```

Downloading to /root/.cache/kagglehub/datasets/bhanupratapbiswas/customer-beh
100%|██████████| 9.94M/9.94M [00:00<00:00, 139MB/s]Extracting files...



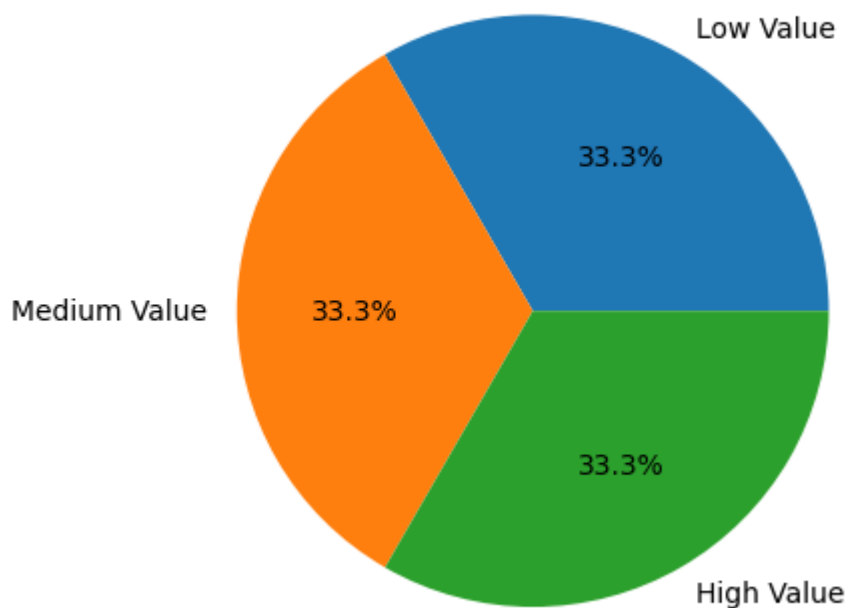
```
df['Customer_Segment'] = pd.qcut(
    df['Total Purchase Amount'],
    q=3,
    labels=['Low Value', 'Medium Value', 'High Value']
)

df['Customer_Segment'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title("Customer Segment Share")
plt.ylabel("")

plt.show()
```

Customer Segment Share



```
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])

monthly_sales = df.groupby(
    df['Purchase Date'].dt.month
)['Total Purchase Amount'].sum()

monthly_sales.plot(kind='line')

plt.title("Monthly Purchase Trend")
plt.xlabel("Month")
plt.ylabel("Total Purchase Amount")

plt.show()
```



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Key Insights and Recommendations

1. High-value customers contribute significantly to the overall purchase amount. AlfidoTech should focus on retaining these customers through loyalty programs and personalized offers.
2. Purchase trends vary across months, indicating seasonal buying behavior.

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